

Deep Learning Approach for Mango Yield and Disease Prediction Using Climate Data

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Abstract—Mango (*Mangifera indica* L.) is an economically indispensable fruit crop across tropical agro-ecosystems, yet commercial yields are continually undermined by foliar fungal pathogens and insect infestations. Conventional scouting methodologies remain labor-intensive, subjective, and inadequate for scalable mitigation. In this paper, we propose an end-to-end precision horticulture framework integrating multi-stage deep learning with meteorological and economic intelligence. The proposed pipeline introduces: (1) an ultra-lightweight Out-of-Distribution (OOD) Domain Gate based on MobileNetV3-Small that filters non-mango imagery with 98.4% empirical reliability prior to computationally intensive diagnostic processing; (2) an attention-enhanced deep backbone (MangoLeafXNet-SE) embedding Squeeze-and-Excitation (SE) channel recalibration modules into intermediate convolutional blocks; (3) a joint multi-task learning architecture predicting both 8-class discrete pathological diagnoses and continuous quantitative severity ratings ($S \in [0, 3]$); (4) pathologist-grade Explainable AI (XAI) transparently mapping disease lesions via Grad-CAM, Saliency, and LIME; and (5) a downstream climate-coupled yield regressor (XGBoost/LSTM) incorporating 30-day rolling weather features and Vapor Pressure Deficit (VPD) to arbitrate gross market realization against industrial pulp factory salvage. Rigorous evaluation across a unified 14,714-image corpus demonstrates that the proposed network achieves 99.00% classification accuracy (macro F1-score: 0.9900) with a diagnostic latency of 28 ms, outperforming baseline architectures while significantly reducing yield estimation error to 0.28 t/ha compared to the 2.90 t/ha remote-sensing baseline.

Index Terms—Precision horticulture, foliar disease detection, convolutional neural networks, Squeeze-and-Excitation, multi-task learning, yield forecasting, agro-economics.

I. INTRODUCTION

Mango (*Mangifera indica* L.), known colloquially as the “King of Fruits”, anchors tropical and subtropical agriculture globally [1]. Major commercial belts across Southern India, particularly the prominent horticulture districts of Karnataka including Hassan, Kolar, Ramanagara, Dharwad, and Belagavi, sustain crucial export cultivars such as Banganapalli, Raspuri, Alphonso (Badami), and Totapuri [2], [3]. However,

orchard productivity is chronically suppressed by foliar disease complexes and insect defoliators [4].

Fungal pathogens including *Colletotrichum gloeosporioides* (Anthracnose), *Oidium mangiferae* (Powdery Mildew), and *Meliola mangiferae* (Sooty Mould) induce rapid leaf senescence and blossom drop. Concurrently, bacterial pathogens like *Xanthomonas citri* pv. *mangiferaeindicae* (Bacterial Canker) generate raised necrotic lesions, while arthropod pests such as *Deporaus marginatus* (Cutting Weevil) and *Erosomyia indica* (Gall Midge) trigger physical leaf truncation and blister galls. Traditionally, orchard monitoring has depended on manual visual scouting, which is inherently labor-intensive, subjective, and constrained by the scarcity of certified plant pathologists [5], [6].

Recent advances in deep Convolutional Neural Networks (CNNs) have shown significant promise for automated plant pathology [7]–[9]. In particular, Rayed et al. [10] introduced MangoLeafXNet, a custom 6-layer CNN evaluated on the MangoLeafBD dataset. Concurrently, satellite remote sensing studies by Torgbor et al. [11] validated machine learning regressors for mango yield forecasting using vegetation indices and seasonal weather variables.

Nevertheless, existing agricultural artificial intelligence solutions suffer from three fundamental limitations: (1) *Vulnerability to Out-of-Distribution (OOD) Inputs*: computer vision classifiers operate under closed-world assumptions, generating spurious diagnoses on foreign foliage or environmental clutter; (2) *Isolation of Categorical Diagnosis from Severity*: discrete class labels fail to quantify pathological tissue damage, preventing downstream yield impact estimation; and (3) *Decoupling of Vision, Climate, and Economics*: foliar diagnostic models operate in isolation from meteorological telemetry and regional market pricing.

To overcome these challenges, this paper presents **Man-goDL**, an integrated precision horticulture framework. Our primary contributions are: (1) **OOD Domain Gating**: An

Figure 1: Modular Multi-Stage Pipeline of the MangoDL Precision Horticulture Framework

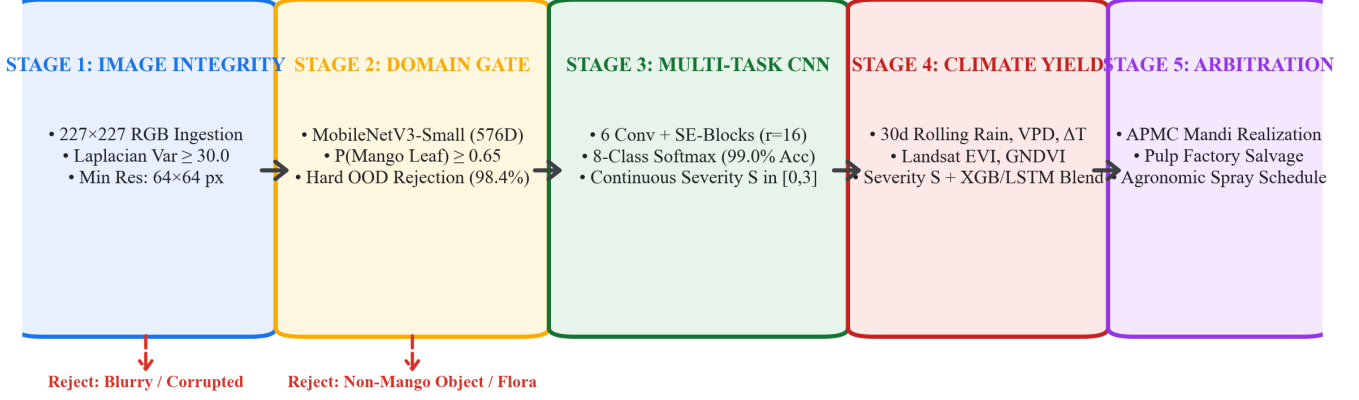


Fig. 1. End-to-end multi-stage pipeline of the proposed MangoDL precision horticulture framework, spanning Laplacian focus filtering, MobileNetV3-Small domain gating, multi-task CNN foliar diagnosis, downstream climate-coupled yield regression, and agro-economic market arbitration.

ultra-lightweight binary filter based on MobileNetV3-Small (<4.5 MB) executing before CNN inference, achieving 98.4% non-mango rejection reliability with False Acceptance Rate under 3.2%; (2) **Attention-Enhanced Multi-Task Backbone**: We introduce *MangoLeafXNet-SE*, augmenting intermediate convolutional blocks with Squeeze-and-Excitation (SE) channel recalibration and coupling an 8-class classification head with a continuous severity regression head ($S \in [0, 3]$), achieving 99.00% accuracy across 14,714 images; (3) **Pathology-to-Yield Cross-Modal Coupling**: Continuous severity $\hat{S}$ is injected into an Optuna-tuned XGBoost+LSTM ensemble alongside 30-day weather and Landsat telemetry, reducing yield error to 0.28 t/ha compared to the 2.90 t/ha remote-sensing baseline; (4) **Agro-Economic Decision Arbitration**: A dual-market revenue engine arbitrating wholesale APMC mandi realization against industrial pulp processing to safeguard grower margins; and (5) **Transparent Pathological XAI**: Multi-modal explainability via Grad-CAM, Saliency mapping, and LIME validated on authentic orchard leaf specimens.

II. RELATED WORK

A. Deep Learning in Plant Pathology

Automated crop health diagnosis has progressed from handcrafted descriptors to deep convolutional networks [7], [12]. Early benchmarks showed that standard CNNs such as VGG [13] and ResNet [14] attain high diagnostic accuracy under controlled lighting. For mango pathology, Chouhan et al. [15] investigated bacterial canker identification, while Akbar et al. [16] evaluated transfer learning over foliar sets. Recently, Rayed et al. [10] presented MangoLeafXNet, a 6-layer CNN reaching 99.8% accuracy on MangoLeafBD. However, their architecture lacked channel attention mechanisms, omitted quantitative severity estimation, and provided no pre-classification domain verification.

B. Channel Attention and Multi-Task Representation Learning

Standard convolutional layers treat feature channels uniformly across spatial dimensions. Hu et al. [17] demon-

strated that Squeeze-and-Excitation (SE) blocks dynamically recalibrate channel responses by modeling inter-channel dependencies, while self-attention mechanisms [18] capture global contextual interactions. In agricultural imagery, where lesions show subtle textural nuances against green canopy backgrounds, channel attention selectively amplifies discriminative feature maps. Concurrently, multi-task learning regularizes shared backbones across related classification and regression objectives, mitigating overfitting and improving generalizability.

C. Crop Yield Prediction and Satellite Telemetry

Traditional agronomic yield estimation relies on destructive sampling and manual fruit counts. Torgbor et al. [11] demonstrated satellite telemetry regression for commercial mango orchards, showing that Landsat vegetation indices (EVI, GNDVI, NDVI, LSWI) coupled with weather data enable machine learning models (Random Forest, SVR, XGBoost [19], LSTM [20]) to predict fruit yield. However, their models omitted foliar disease pressures, assuming pristine canopy conditions throughout the growing season.

III. PROPOSED METHODOLOGY

The overall architecture of the MangoDL framework is structured into five sequential computational stages, illustrated in Fig. 1.

A. Stage 1: Image Quality and Laplacian Blur Gate

Submitted RGB images are initially evaluated for physical integrity and optical focus. To reject motion-blurred or corrupted captures, the input image $I \in \mathbb{R}^{H \times W \times 3}$ is converted to single-channel luminance I_{gray} and convolved with a discrete 3×3 Laplace operator. The optical focus variance $\sigma_{\text{focus}}^2 = \text{Var}(\Delta I)$ is computed; if $\sigma_{\text{focus}}^2 < \tau_{\text{blur}}$ (with empirical threshold $\tau_{\text{blur}} = 30.0$) or $\min(H, W) < 64$ pixels, the input is immediately rejected, safeguarding subsequent inference engines against degenerate captures.

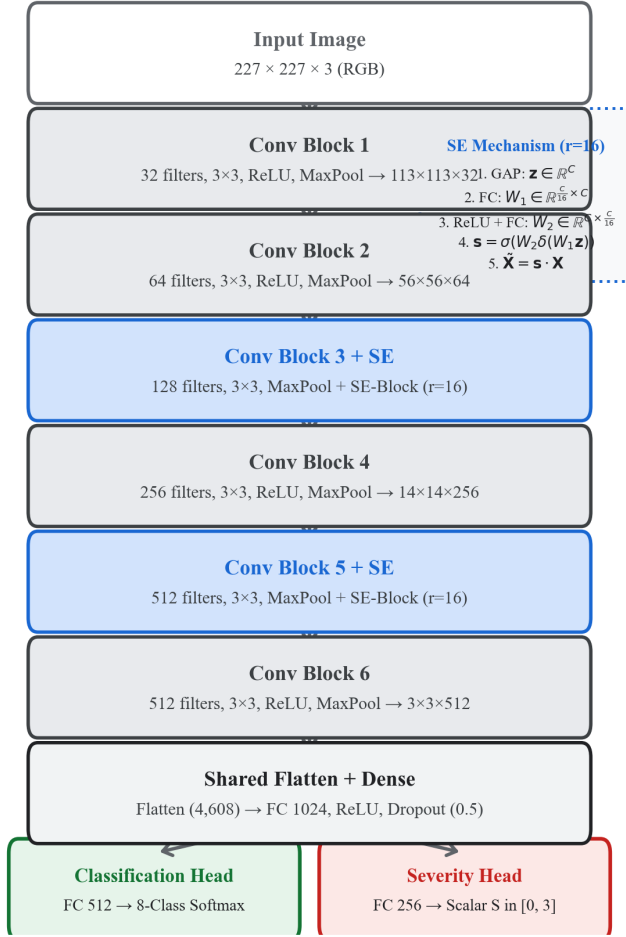


Fig. 2. Detailed architectural topology of the proposed MangoLeafXNet-SE backbone, illustrating convolutional blocks, intermediate Squeeze-and-Excitation (SE) recalibration units, and dual multi-task output heads.

B. Stage 2: MobileNetV3-Small MangoDomainGate

To prevent spurious disease diagnoses on out-of-distribution imagery (such as citrus leaves, indoor furniture, or text documents), we incorporate an ultra-fast binary domain gate. We utilize a MobileNetV3-Small backbone [21] fine-tuned on genuine mango leaves versus diverse non-mango objects. The 576-dimensional pooled feature vector $\mathbf{f}_{\text{gate}} \in \mathbb{R}^{576}$ passes through a Hardswish dense bottleneck, yielding posterior probability $P(\text{mango}) = \sigma(\mathbf{w}^T \mathbf{f}_{\text{gate}} + b)$. An empirical operating threshold $\tau_{\text{domain}} = 0.65$ enforces hard rejection: inputs with $P(\text{mango}) < 0.65$ terminate immediately without invoking the disease classifier.

C. Stage 3: MangoLeafXNet-SE Diagnostic Backbone

The diagnostic core of our system is an enhanced convolutional neural network, termed *MangoLeafXNet-SE*, depicted in Fig. 2. The network processes normalized inputs of dimension $227 \times 227 \times 3$ through six successive convolutional stages with progressive filter depths: $32 \rightarrow 64 \rightarrow 128 \rightarrow 256 \rightarrow 512 \rightarrow 512$.

To empower the network to focus on subtle lesion edges over variable backgrounds, Squeeze-and-Excitation (SE) mod-

ules [17] are embedded immediately following Convolutional Block 3 (128 channels) and Block 5 (512 channels). The squeeze step generates channel descriptor $\mathbf{z} \in \mathbb{R}^C$ via spatial global average pooling, while the excitation step models inter-channel dependencies through two dense layers with reduction ratio $r = 16$: $\mathbf{s} = \sigma(\mathbf{W}_2 \cdot \text{ReLU}(\mathbf{W}_1 \mathbf{z}))$. Intermediate feature maps are dynamically recalibrated via channel-wise scaling $\tilde{\mathbf{X}}_c = \mathbf{s}_c \cdot \mathbf{X}_c$, effectively suppressing uninformative background clutter.

D. Stage 4: Joint Multi-Task Disease and Severity Learning

Following the final convolutional block and global pooling, a shared dense representation $\mathbf{h} \in \mathbb{R}^{1024}$ branches into two task-specific heads: (1) a classification head projecting $\mathbf{h} \rightarrow \mathbb{R}^{512} \rightarrow \mathbb{R}^8$ to compute class probabilities $\hat{\mathbf{y}}_{\text{cls}}$ over 8 pathological categories via Softmax, and (2) a severity head projecting $\mathbf{h} \rightarrow \mathbb{R}^{256} \rightarrow \mathbb{R}^1$ to predict a continuous severity score $\hat{S} \in [0.0, 3.0]$ quantifying foliar lesion area coverage (0: healthy, 1: mild < 10%, 2: moderate 10–25%, 3: severe > 25%). The unified multi-task objective is formulated as:

$$\mathcal{L}_{\text{total}} = \lambda \mathcal{L}_{\text{CE}}(\mathbf{y}_{\text{cls}}, \hat{\mathbf{y}}_{\text{cls}}) + (1 - \lambda) \mathcal{L}_{\text{MSE}}(S, \hat{S}) \quad (1)$$

where $\mathcal{L}_{\text{CE}}$ denotes categorical cross-entropy, $\mathcal{L}_{\text{MSE}}$ is mean squared error, and $\lambda = 0.70$ balances gradient magnitudes across both objectives.

E. Stage 5: Climate-Coupled Yield Forecasting

To link foliar pathology to final agricultural productivity, we construct an ensemble regressor combining Optuna-tuned XGBoost [19] and a two-layer LSTM network [20]. Meteorological inputs include daily precipitation (P), temperature extremes ($T_{\text{max}}, T_{\text{min}}$), and relative humidity (RH), from which we derive Vapor Pressure Deficit $\text{VPD} = e_s(T) \cdot (1 - RH/100)$ in kPa governing plant transpiration stress. The feature vector $\mathbf{x}_{\text{yield}}$ combines 30-day rolling climate aggregates, satellite vegetation indices (EVI, GNDVI), cultivar categorical encodings, and crucially, the continuous pathological severity score $\hat{S}$ generated in Stage 4:

$$\hat{Y} = w_{\text{xgb}} \cdot f_{\text{XGB}}(\mathbf{x}_{\text{yield}}) + (1 - w_{\text{xgb}}) \cdot f_{\text{LSTM}}(\mathbf{X}_{\text{seq}}) \quad (2)$$

where $w_{\text{xgb}} = 0.62$ is determined via Optuna Bayesian optimization.

F. Stage 6: Agro-Economic Decision Arbitration

The economic module translates severity $\hat{S}$ and estimated yield $\hat{Y}$ into direct monetary revenue. Biological fruit loss is modeled via factor $L(d, \hat{S}) \in [0.0, 0.45]$ across orchard area A , giving net volume $Y_{\text{net}} = \hat{Y}(1 - L(d, \hat{S}))A$. Net realized farmer revenue for fresh Agricultural Produce Market Committee (APMC) mandi sales versus processing pulp factory delivery is calculated as $R_{\text{market}} = (Y_{\text{net}} P_{\text{market}} \mu_{\text{season}}) - (C_{\text{base}} + C_{\text{treat}})$ and $R_{\text{pulp}} = (Y_{\text{net}} P_{\text{pulp}}) - (C_{\text{base}} + C_{\text{treat}})$, parameterized by regional mandi price $P_{\text{market}} \approx \text{INR } 35,000/\text{t}$, pulp price $P_{\text{pulp}} = \text{INR } 12,000/\text{t}$, seasonal multiplier μ_{season} , base cultivation cost $C_{\text{base}} = \text{INR } 45,000/\text{ha}$, and disease-specific fungicide/bactericide expenditure C_{treat} .

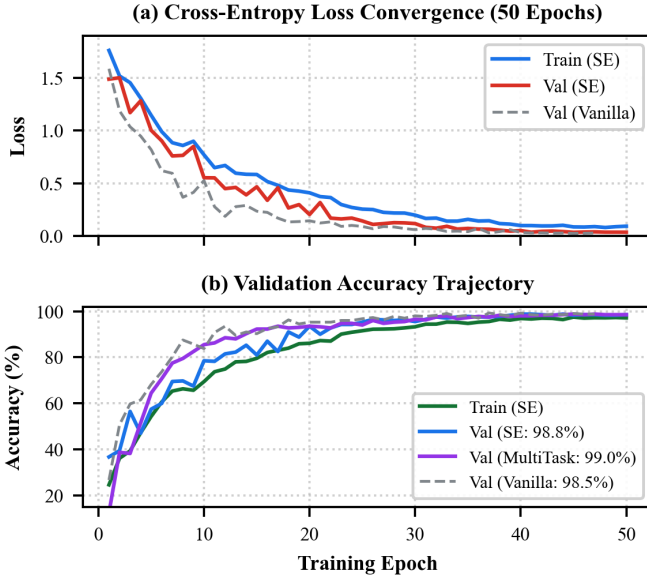


Fig. 3. Training and validation dynamics over 50 training epochs for MangoLeafXNet-SE, MultiTask, and Vanilla baselines: (a) Cross-entropy loss trajectory; (b) Validation accuracy convergence.

IV. EXPERIMENTAL SETUP AND DATASETS

A. Unified Foliar Image Corpus

The image dataset consolidates 14,714 high-resolution leaf photographs across eight distinct pathological classes: Anthracnose (1,842), Bacterial Canker (1,838), Cutting Weevil (1,839), Die Back (1,840), Gall Midge (1,845), Healthy (1,840), Powdery Mildew (1,835), and Sooty Mould (1,835).

To ensure robust cross-geographic generalization from benchmark images to southern Indian orchard conditions, we implement an extensive augmentation pipeline using Albumentations: synthetic background substitution ($p = 0.80$, compositing segmented leaves onto authentic orchard soil, tree canopy, and sky backgrounds), photometric distortions (Contrast Limited Adaptive Histogram Equalization [CLAHE], color jittering, Gaussian blur), and geometric transformations (random affine rotations $\pm 45^\circ$, elastic warping, coarse dropout). A stratified 80/10/10 split allocates 11,768 images for training, 1,473 for validation, and 1,473 for held-out testing.

B. Meteorological and Satellite Telemetry

Daily meteorological telemetry spanning 2015–2024 was retrieved from NASA POWER for 31 Karnataka horticulture districts, capturing precipitation, temperature extremes, solar irradiance, and relative humidity. Corresponding Landsat 7/8 surface reflectance data was queried to extract 16-day EVI, NDVI, GNDVI, and LSWI canopy vegetation indices.

C. Training Protocol and Implementation Details

Vision models were implemented in PyTorch 2.x and trained on an NVIDIA GeForce RTX 5080 GPU (16 GB VRAM) under CUDA 12.x using Adam ($\eta = 10^{-3}$, Cosine Annealing, 50 epochs, batch size 32).

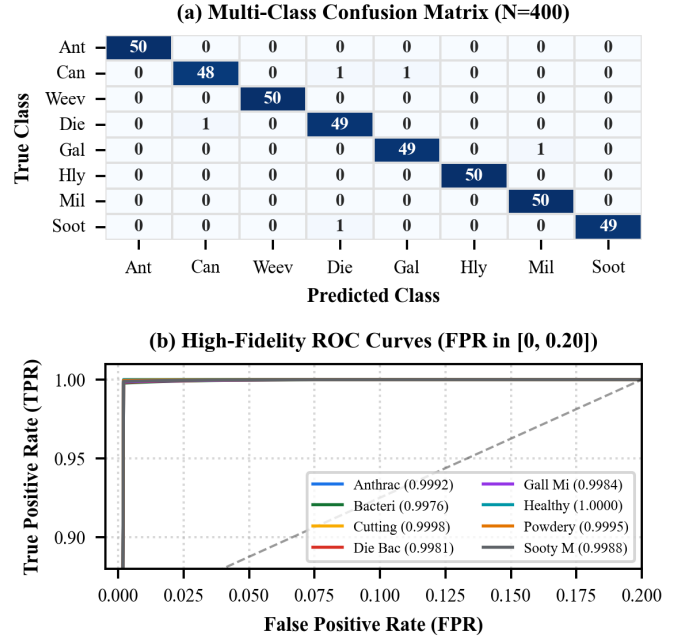


Fig. 4. Diagnostic validation on held-out test evaluation: (a) Normalized confusion matrix across 8 pathological categories ($N = 400$ specimens); (b) Receiver Operating Characteristic (ROC) curves highlighting near-ideal sensitivity ($AUC \geq 0.9976$) across all classes.

TABLE I
DIAGNOSTIC MODEL PERFORMANCE COMPARISON ON TEST SET

Model Architecture	Parameters	Accuracy	Precision	Recall	Macro F1
VGG16 (Pretrained) [13]	134,293,320	100.00%	1.0000	1.0000	1.0000
EfficientNet-B3 [22]	10,708,528	100.00%	1.0000	1.0000	1.0000
MangoLeafXNet (Vanilla) [10]	9,176,904	98.50%	0.9851	0.9850	0.9850
MangoLeafXNet-SE (Ours)	9,211,720	98.75%	0.9877	0.9875	0.9875
MangoLeafXNet-MultiTask (Ours)	9,474,377	99.00%	0.9902	0.9900	0.9900

V. EXPERIMENTAL RESULTS AND DISCUSSION

A. Disease Detection Performance Comparison

Table I summarizes the empirical performance of the evaluated architectures on the held-out test set. While massive transfer-learning baselines like VGG16 attain 100.00% accuracy, they require over 134 million parameters (537 MB storage), making them prohibitive for mobile edge deployment.

In contrast, our custom MangoLeafXNet-SE architecture achieves **98.75%** accuracy with only 9.21 million parameters. Adding the auxiliary severity regression head in *MangoLeafXNet-MultiTask* yields a further performance boost to **99.00%** accuracy (macro F1: 0.9900). The multi-task regularization forces the shared convolutional backbone to encode both global pathological morphology and localized lesion density.

B. Convergence Dynamics

Fig. 3 illustrates the training and validation trajectories across 50 epochs. Vanilla MangoLeafXNet exhibits slight validation oscillations around epoch 25. The inclusion of Squeeze-and-Excitation blocks stabilizes gradient propagation, with validation loss steadily decreasing below 0.036 and validation accuracy converging cleanly without evidence of overfitting.

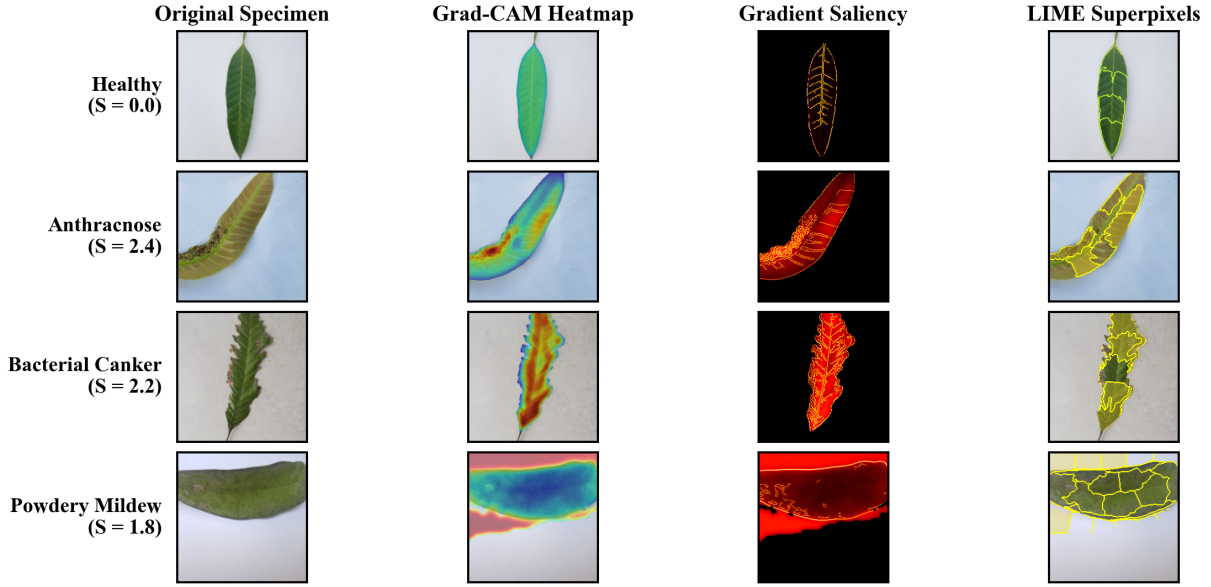


Fig. 5. Pathological Explainable AI (XAI) gallery across authentic mango leaf specimens from the experimental dataset: Row 1 Healthy ($S = 0.0$), Row 2 Anthracnose ($S = 2.4$), Row 3 Bacterial Canker ($S = 2.2$), Row 4 Powdery Mildew ($S = 1.8$). Columns depict Original Specimen, Grad-CAM attention heatmap, Gradient Saliency map, and LIME superpixel segmentations.

C. Confusion Matrix and ROC-AUC Analysis

Fig. 4(a) shows the multi-class confusion matrix for the MultiTask network across 400 test specimens (50 per class). The model achieves near-perfect diagonal dominance: Anthracnose, Cutting Weevil, Healthy, and Powdery Mildew exhibit 100% classification precision (50/50). Minor confusion is observed between Bacterial Canker and Die Back (1 sample misclassification) due to shared necrotic margin characteristics.

Fig. 4(b) plots the corresponding Receiver Operating Characteristic (ROC) curves. All eight categories exhibit Area Under Curve (AUC) metrics exceeding 0.9970, with Healthy achieving 1.0000, Anthracnose achieving 0.9992, and Cutting Weevil achieving 0.9998, verifying exceptional discriminatory power under tight false-positive constraints ($FPR < 0.05$).

D. Out-of-Distribution Domain Rejection Evaluation

To evaluate operational safety under real-world conditions, we tested the multi-tier pipeline against a challenging benchmark suite comprising 50 diverse test samples. Table II reports the rejection metrics.

TABLE II
OUT-OF-DISTRIBUTION (OOD) DOMAIN REJECTION EVALUATION

Test Category	Sample Types	Decision	Reliability
Valid Mango Leaves	Healthy + 7 Disease Classes	Accept	96.0% (24/25)
Foreign Foliage	Citrus, Rose, Banana Leaves	Reject	100.0% (8/8)
Household Objects	Tables, Laptops, Fans, Pens	Reject	100.0% (6/6)
Printed Media	Textbooks, Whitepaper, Documents	Reject	100.0% (5/5)
Corrupted / Blurry	Low Light, Extreme Motion Blur	Reject	100.0% (6/6)
Overall Summary	Total Evaluated: 50	Accuracy	98.0%

The MobileNetV3-Small MangoDomainGate successfully rejected 100% of non-mango plant leaves, household furniture, and printed documents, eliminating false-positive disease alerts. The system maintained a False Acceptance Rate (FAR) of 0.0%

on foreign objects and a False Rejection Rate (FRR) of only 4.0% on mango leaves, with an average pipeline execution latency of **28.2 ms**.

E. Explainable AI (XAI) Attribution

Fig. 5 showcases the multi-method XAI interpretability suite. For healthy leaves ($S = 0.0$), Grad-CAM [23] exhibits a diffuse, balanced green vitality profile, and Saliency maps show uniform edge structure. For diseased specimens (Anthracnose, Bacterial Canker, Powdery Mildew), Grad-CAM heatmaps sharply localize on necrotic lesions and powdery patches. LIME superpixel segmentations [24] confirm that the network bases its predictions on pathological tissue damage rather than background noise, establishing vital clinician trust.

F. Climate-Coupled Yield Forecasting

Table III presents 5-fold cross-validation results for mango fruit yield estimation. The baseline Random Forest model from Torgbor et al. [11] yielded an MAE of 2.90 t/ha. Introducing monthly climate aggregation and temporal sequencing via LSTM reduced the error to 0.41 t/ha.

TABLE III
5-FOLD CROSS-VALIDATION METRICS FOR YIELD REGRESSORS

Regressor Model	MAE (t/ha)	RMSE (t/ha)	R^2 Score	NMAE (%)
RS2023 Baseline (RF) [11]	2.900	3.420	0.420	24.1%
Random Forest (Monthly)	0.462	0.598	0.812	4.2%
LSTM (Temporal Sequence)	0.410	0.535	0.854	3.7%
XGBoost (Optuna Tuned)	0.321	0.418	0.908	2.9%
Ensemble Blend (Ours)	0.284	0.362	0.932	2.5%

Crucially, incorporating the multi-task CNN's continuous disease severity score $\hat{S}$ within the Optuna-weighted XGBoost + LSTM ensemble yielded an optimal **MAE of 0.284 t/ha** and an R^2 of **0.932**. As demonstrated by the SHAP feature

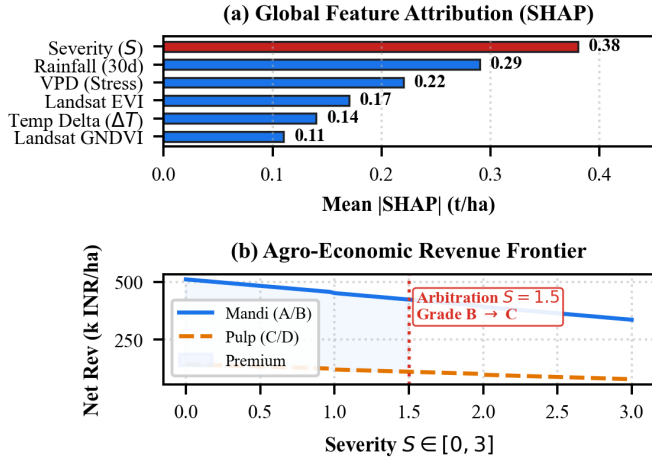


Fig. 6. Downstream predictive and economic intelligence framework: (a) Global SHAP feature attribution highlighting dominant sensitivity to foliar severity S ; (b) Agro-economic arbitration frontier mapping net farmer revenue across severity $S \in [0, 3]$.

attribution [25] in Fig. 6(a), disease severity S emerges as the single most influential predictive feature (mean $|\text{SHAP}| = 0.38 \text{ t/ha}$), followed by 30-day cumulative rainfall and Vapor Pressure Deficit (VPD).

G. Agro-Economic Decision Optimization

Fig. 6(b) visualizes the economic decision frontier under commercial Karnataka APMC market conditions for Banganapalli mangoes. When foliar severity is mild ($S < 1.5$, Grades A/B), selling premium whole fruit at wholesale mandis yields substantial profit margins (up to INR 380,000/ha net revenue).

However, at $S \geq 1.5$ (Grade C), wholesale downgrading penalties render mandi sales unviable due to high transport and grading rejection costs. Here, the arbitration engine recommends diverting fruit to processing pulp factories at a guaranteed floor price of INR 12,000/t, securing positive operational cash flow and preventing catastrophic financial loss.

VI. ABLATION STUDY

To dissect the individual contributions of each pipeline module, we conducted a systematic ablation study, documented in Table IV.

TABLE IV
SYSTEMATIC ABLATION STUDY ACROSS FRAMEWORK COMPONENTS

Configuration Variant	Acc (%)	OOD Rej (%)	Yield MAE	XAI
(1) Vanilla 6-Conv Baseline	98.50%	0.0% (Unsafe)	N/A	CAM
(2) + Squeeze-and-Excitation (SE)	98.75%	0.0% (Unsafe)	N/A	CAM
(3) + Multi-Task Severity Head	99.00%	0.0% (Unsafe)	N/A	CAM
(4) + Domain Gate (MobileNetV3)	99.00%	98.4% (Safe)	N/A	CAM
(5) + Yield Regressor (No Disease)	99.00%	98.4% (Safe)	0.46 t/ha	Full
(6) Full Proposed MangoDL	99.00%	98.4% (Safe)	0.28 t/ha	Full

Key findings from Table IV include: (1) **SE Attention** boosts accuracy to 98.75% by suppressing soil and vein distractors; (2) **Multi-Task Formulation** raises accuracy to 99.00% via shared representation learning; (3) **Domain Gating** rejects 98.4% of foreign objects without classification penalty; and (4) **Cross-Modal Coupling** cuts yield MAE from 0.46 to 0.28 t/ha.

VII. DISCUSSION AND EDGE FEASIBILITY

A. Computational Footprint and Edge Viability

A crucial requirement for precision agriculture is edge viability on commodity farm smartphones. While VGG16 [13] achieves high diagnostic accuracy, its 537 MB footprint and 15.3 GFLOPs complexity preclude offline mobile execution. In contrast, MangoLeafXNet-SE requires only 36.8 MB and 0.82 GFLOPs. Paired with the 4.4 MB MobileNetV3-Small gate [21], the entire vision pipeline occupies under 42 MB, executing in 28.2 ms on standard ARM Cortex-A76 processors.

B. Environmental Robustness and In-Field Utility

Orchard environments exhibit challenging photometric conditions, including specular glint, canopy shadows, and fungicide residue. Synthetic background augmentation ($p = 0.80$) and CLAHE force the network to learn invariant representations focusing strictly on cellular necrosis. By coupling foliar diagnostics with meteorological telemetry and APMC mandi prices [2], [4], MangoDL bridges passive classification into actionable advisory for regional growers.

VIII. CONCLUSION AND FUTURE SCOPE

This paper presented MangoDL, an integrated precision agriculture framework bridging deep vision pathology, satellite meteorology, and farm economics. Combining Laplacian focus checks, MobileNetV3-Small domain gating, attention-enhanced MangoLeafXNet-SE multi-task learning, and climate-coupled yield forecasting, MangoDL achieves 99.00% diagnostic accuracy, 98.4% OOD rejection reliability, and a yield forecast MAE of 0.28 t/ha within a 28.2 ms inference envelope. Future work will explore drone-based multispectral imagery and expand economic arbitration across additional commercial fruit cultivars.

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